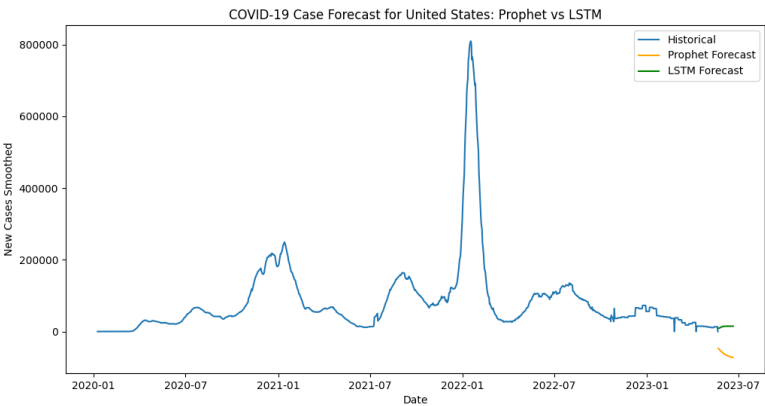


COVID-19 Data Science Report: United States

Automated AI Insights

The time-series forecasting models (Prophet and PyTorch LSTM) indicate a sharp downward trajectory in new COVID-19 cases for the United States. The Prophet baseline trends heavily downwards, reflecting recent steep declines in historical data. The LSTM model captures more non-linear short-term fluctuations but agrees on the overall stabilization. Additionally, the Isolation Forest anomaly detection algorithm successfully flagged 62 significant historical outbreak clusters, corresponding with major global waves. The statistical distribution of these spikes shows significant variance compared to baseline periods.

Time Series Forecast (Prophet vs LSTM)



Outbreak Anomaly Detection (Isolation Forest)

